

Design Proposal of Mach-Zehnder Interferometer

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Abstract: Design and simulation of waveguides for Mach-Zehnder interferometer.

1. Introduction

Mach-Zehnder interferometers (MZIs) are fundamental building blocks in integrated photonic circuits, enabling optical interference through controlled splitting, propagation, and recombination of guided optical signals. The spectral response of an MZI is determined by the relative phase accumulated between its two arms, which can be controlled through the path-length difference, ΔL . This design proposal focuses on the layout and characterization of waveguides and MZIs operating in cband TE mode at 1550nm. The proposed structures include a symmetric reference MZI and asymmetric MZIs with $\Delta L = 0, 25, 50, 75$, and $100 \mu\text{m}$. Here, we use TE 1550nm grating couplers and have variations of the MZIs with Y-branches and directional couplers. In the simulations, different waveguide widths are considered to characterize their influence on waveguide performance.

2. Theory

2.1. Basic Theory of the Photonic Components

2.1.1. Mach-Zehnder Interferometer

In Mach-Zehnder interferometer (MZI) the input optical field is divided by waveguide splitters, resulting in two arms, and recombined at the output. During propagation a relative phase difference is accumulated. In an MZI with an optical path-length difference ΔL , a relative phase difference can be approximated as

$$\Delta\phi = \frac{2\pi n_{\text{eff}} \Delta L}{\lambda}, \quad (1)$$

where n_{eff} is the effective modal index and λ is the operating wavelength. The output transmission varies periodically with wavelength due to constructive and destructive interference. For an asymmetric MZI, the free spectral range (FSR) is approximated with n_g the group index as [1]:

$$\text{FSR} \approx \frac{\lambda^2}{n_g \Delta L}, \quad (2)$$

3. Modeling and Simulations

3.1. waveguide

In our design, we use strip waveguide with 500nm width and 220nm height. The simulated waveguide mode profiles are presented in the following figures. In this design, the height of the waveguide was set at 220nm, a common value. The waveguide width was varied

The effective index versus wavelength of the simulated waveguide is shown in Figure 5 while the group index versus waveguide is seen in Figure 3.

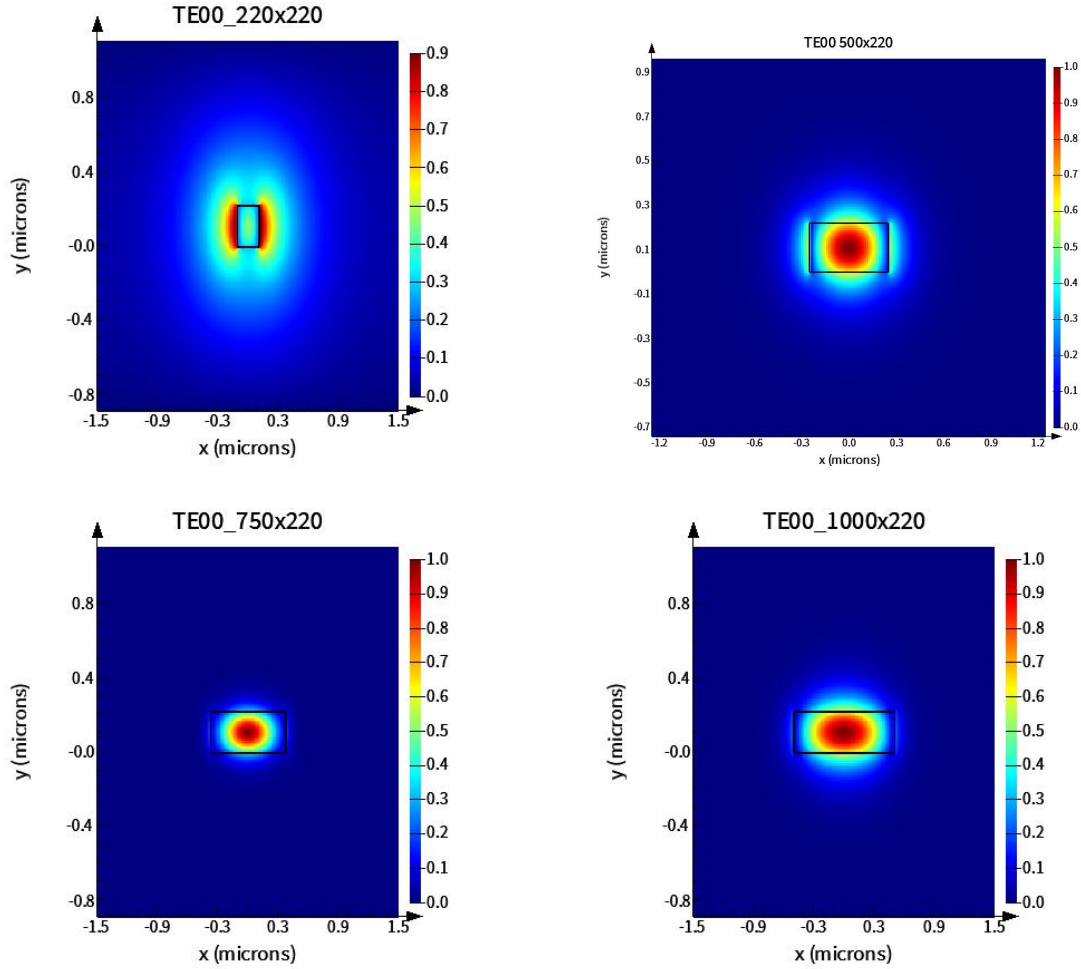


Fig. 1. Different TE₀₀ modes with changing width.

width (microns)	effective index
1000	2.7445
750	2.6662
500	2.4445
220	1.5019

TABLE I

VARIED WIDTHS USED TO SIMULATE THE DIFFERENT TE MODES AND THEIR RESPECTIVE EFFECTIVE INDEX.

Using Lumerical the compact model of the waveguide is $2.44435 - 1.13055 * (\lambda - 1.55) - 0.0332067 * (\lambda - 1.55)^2$
this can be shown in 4 below.

3.2. Mach-Zehnder Interferometer

The schematic used to model the Mach-Zehnder Interferometer is shown in
The free spectral range simulations did not work.

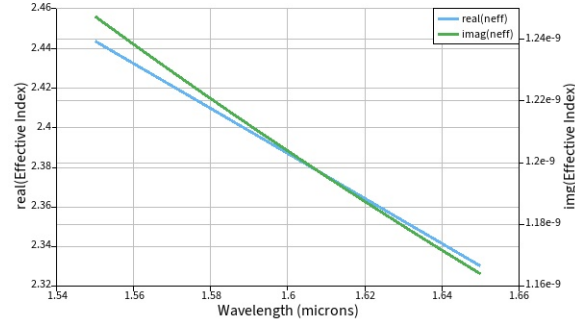


Fig. 2. Effective index versus wavelength.

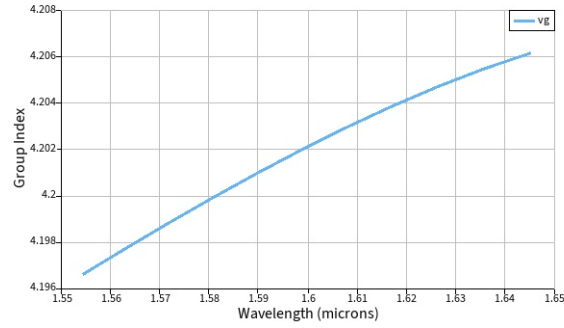


Fig. 3. Group index versus wavelength.

4. Layout

The proposed layout consists of a systematic set of Mach-Zehnder interferometer (MZI) structures designed for optical characterization at 1550 nm, see Fig. 6. The layout includes strip waveguides with a widths of 500 nm together with TE 1550 nm grating couplers (GCs), Y-branches or direct-coupler strucutres.

- **Grating couplers:** Grating couplers are incorporated for optical input/output coupling at 1550 nm. The pitch between grating couplers belonging to different MZI structures is approximately 200 μm . We are using the $GC_{TE1550}degOxide_B B$ from the PDK.
- **TE 1550 nm couplers:** Directional coupling structures designed for TE-polarized operation

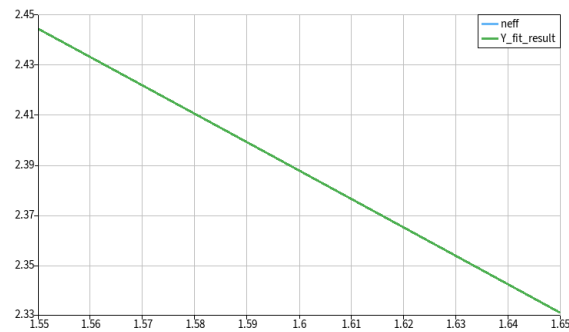


Fig. 4. compact fit model.

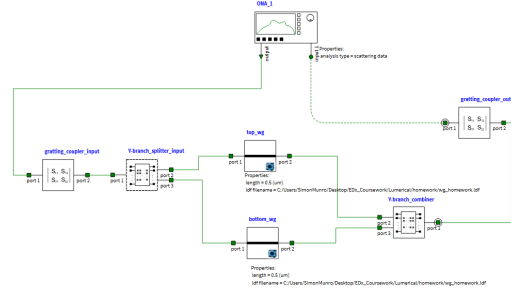


Fig. 5. diagram of setup on interconnect.

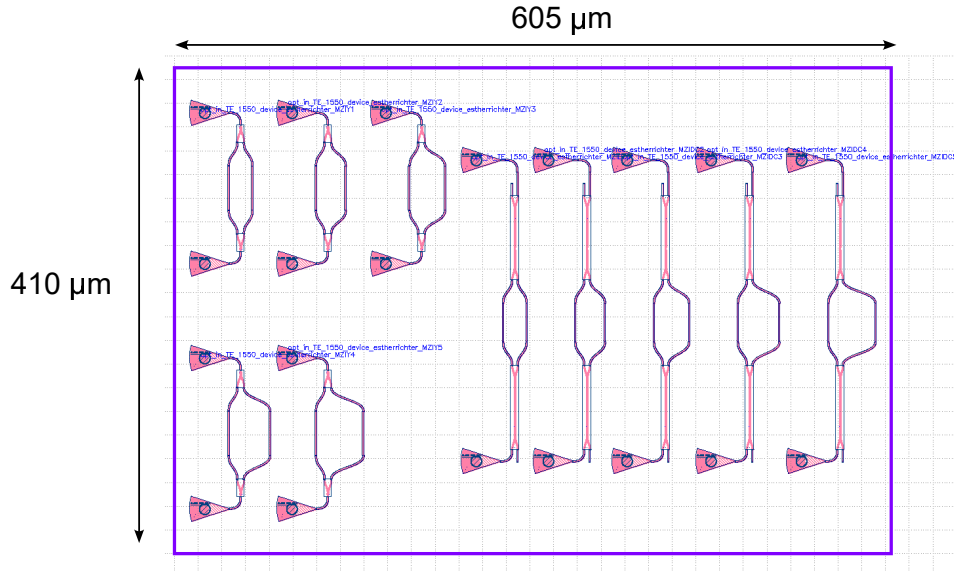


Fig. 6. Design of MZIs with TE grating couplers.

at 1550 nm are included for characterization of the coupling response. We are using the *ebeam_{bdc}e1550* structure from the PDK.

- **Y-branches:** Y-branches and directional couplers are used to form the interferometric structures to independently evaluate the performance of the corresponding components. We are using *ebeam_{y1550}* from the PDK.
- **Mach-Zehnder interferometers:** A series of MZIs with different path-length differences, ΔL , both using Y-branches and directional couplers is included:

$$\Delta L = 0, 25, 50, 75, 100 \mu\text{m}.$$

The $\Delta L = 0 \mu\text{m}$ structure corresponds to a symmetric MZI as a reference structure, while the remaining structures introduce increasing path-length asymmetry.

Overall, the layout provides a structured test matrix combining different coupling structures, and MZI path-length differences. This enables systematic characterization of the individual building blocks and investigation of the dependence of MZI spectral response on waveguide geometry and path-length difference.



References

References

[1] J. N. Caspers *et al.*, “Mach-zehnder interferometric engineering in silicon optical modulators,” *Applied Optics*, vol. 62, pp. 7387–7397, 2023.

